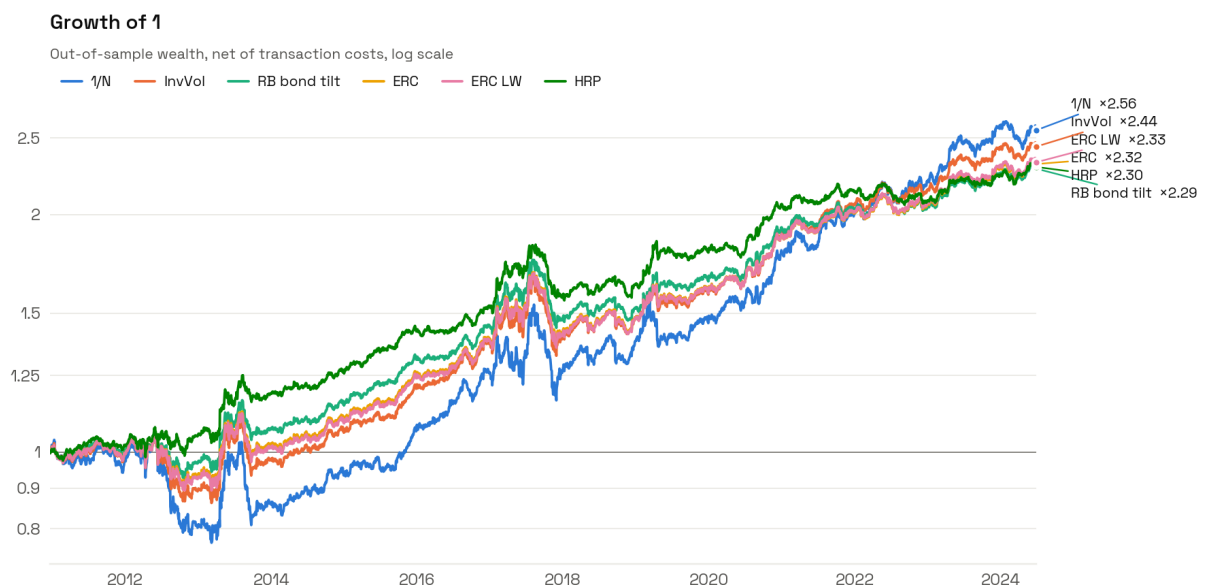


Risk-based portfolio allocation

Methodology, proofs
and performance measurement

This document describes the methods implemented in the `portopt` library: covariance estimation, risk contributions, $1/N$, inverse volatility, risk budgeting, ERC and HRP allocations, walk-forward backtesting, performance and risk metrics, and significance tests. Each method is presented through its intuition, its theory with proofs, a worked example, practitioner's tips and where it lives in the code.



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Code	<code>portopt/ Figures docs/make_figures.py</code>

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Notation

N, T	number of assets, number of observations
$R_t \in \mathbb{R}^N$	simple returns of the assets at date t (close $t - 1$ to close t)
$w \in \mathbb{R}^N$	portfolio weights, $\mathbf{1}^\top w = 1$, $w \geq 0$ (long-only)
$\Sigma, \sigma_i, \rho_{ij}$	covariance matrix, volatilities $\sigma_i = \sqrt{\Sigma_{ii}}$, correlations
$\sigma_p(w)$	portfolio volatility, $\sqrt{w^\top \Sigma w}$
S	sample covariance
P	number of periods per year ($P = 252$ for daily data)
L, H	in-sample window length, holding period (backtest)
$\bar{R}, \hat{\sigma}$	sample mean and standard deviation (ddof=1)

Figures are produced on simulated data from `portopt.data.simulate_returns` (10 multi-asset ETFs, 15 years, stress regimes and fat tails) with `python docs/make_figures.py`.

1 Data and returns

1.1 Simple and log returns

Definition 1.1. For a price P_t , the simple return and the log return are

$$R_t = \frac{P_t}{P_{t-1}} - 1, \quad r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t).$$

Their aggregation properties are opposite.

	Cross-sectional aggregation (across assets)	Time aggregation
Simple R	$R_{p,t} = \sum_i w_i R_{i,t}$ exact	$\prod_t (1 + R_t) - 1$
Log r	$r_{p,t} \neq \sum_i w_i r_{i,t}$	$\sum_t r_t$ exact

Proposition 1.2. *The simple return of a portfolio is the weighted average of the simple returns of its assets, $R_{p,t} = w^\top R_t$, where w are the value weights at $t - 1$. The property does not hold for log returns.*

Proof. The portfolio value is $V_t = \sum_i n_i P_{i,t}$ for holdings n_i fixed over the period. Then

$$\frac{V_t}{V_{t-1}} = \sum_i \frac{n_i P_{i,t-1}}{V_{t-1}} \cdot \frac{P_{i,t}}{P_{i,t-1}} = \sum_i w_i (1 + R_{i,t}),$$

hence $R_{p,t} = w^\top R_t$. For log returns, $r_{p,t} = \ln \sum_i w_i e^{r_{i,t}}$, and by strict concavity of the logarithm (Jensen), $\ln \sum_i w_i e^{r_i} > \sum_i w_i r_i$ as soon as the r_i are not all equal. ■

Consequence. Portfolio construction and backtesting are always done in simple returns. Log returns are for statistical modelling (more symmetric, additive over time). Useful relation: $r = \ln(1 + R) = R - R^2/2 + O(R^3)$, a negligible gap at daily frequency but not at annual frequency.

1.2 Time alignment

In `portopt`, R_t dated t is the return from the close of $t - 1$ to the close of t . A weight decided with the information available at the close of $t - 1$ earns R_t . This alignment is what rules out look-ahead bias in the backtest (section 5).

1.3 Cleaning

`clean_prices` applies three rules.

1. **Limited forward-fill** (`ffill_limit=5`): fills exchange holidays when mixing markets. Beyond that, a gap is genuinely missing data.
2. **Never back-fill**: `bfill` copies a future price into the past, which is look-ahead bias in disguise.
3. **Leading NaNs are kept**: an ETF launched in 2012 has NaNs before. The backtester only invests in it once it has a full in-sample window. Dropping “incomplete” assets upfront would keep only the survivors: *survivorship bias*.

1.4 Adjusted prices

`download_prices(auto_adjust=True)` returns closes adjusted for splits and dividends, hence *total return* series. On raw prices, every ex-dividend date looks like a loss: a bond ETF would artificially lose 3 to 4% a year.

In the code

```
portopt/data.py: download_prices, clean_prices, compute_returns, resample_returns,
load_returns, simulate_returns.
```

2 Estimating the covariance

Every method except $1/N$ consumes Σ . The quality of the allocation is bounded by the quality of this estimator.

2.1 Sample covariance

$$S = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})(R_t - \bar{R})^\top.$$

S is unbiased but noisy as soon as N/T is not small. It has $N(N+1)/2$ parameters: 55 for 10 assets, 5,050 for 100 assets. With $T = 252$, estimating 5,050 parameters from 25,200 numbers makes no sense. If $T \leq N$, S has rank at most $T-1$: it is singular and cannot be inverted.

Theorem 2.1 (Marchenko and Pastur, 1967). *If the R_t are iid with covariance I_N and $N, T \rightarrow \infty$ with $q = N/T \in (0, 1]$, the empirical distribution of the eigenvalues of S converges to a law supported on*

$$\lambda \in [(1 - \sqrt{q})^2, (1 + \sqrt{q})^2].$$

With $q = 0.5$, the eigenvalues spread between 0.09 and 2.91 while they are all equal to 1. **Large eigenvalues are overestimated, small ones underestimated.** Yet any method that inverts Σ (minimum variance, Markowitz) puts the most weight on the lowest-variance directions, exactly where the estimate is most wrong: this is the *error maximization* of Michaud (1989).

2.2 Ledoit-Wolf

The idea is to combine S (unbiased, high variance) with a structured target (biased, zero variance). The optimal bias-variance trade-off has a closed form (Ledoit and Wolf, 2004).

Definition 2.2 (Ledoit-Wolf estimator). Let S be the sample covariance normalized by T^1 and $\|A\|^2 = \text{tr}(AA^\top)/N$ the normalized Frobenius norm. Let $\mu = \text{tr}(S)/N$ and

$$d^2 = \|S - \mu I\|^2, \quad \bar{b}^2 = \frac{1}{T^2} \sum_{t=1}^T \|x_t x_t^\top - S\|^2, \quad \delta^* = \frac{\min(\bar{b}^2, d^2)}{d^2},$$

where $x_t = R_t - \bar{R}$. The estimator is $\hat{\Sigma} = \delta^* \mu I + (1 - \delta^*) S$.

Interpretation. d^2 measures how far S is from the target (dispersion of its eigenvalues). $\bar{b}^2$ measures how uncertain S is (sampling variance estimated from the observations themselves).

¹The paper and the code use the $1/T$ normalization for S in the shrinkage formula; the difference with $1/(T-1)$ is of order $1/T$.

δ^* is the share of the observed dispersion attributable to noise: the smaller T relative to N , the stronger the shrinkage.

Properties. For $\delta^* > 0$, $\hat{\Sigma}$ is positive definite. Eigenvalues are pulled toward μ *without changing the eigenvectors*, and the condition number improves.

Lemma 2.3 (fast computation of $\bar{b}^2$). *With $S = \frac{1}{T} \sum_t x_t x_t^\top$,*

$$\sum_{t=1}^T \|x_t x_t^\top - S\|_F^2 = \sum_{t=1}^T (x_t^\top x_t)^2 - T \|S\|_F^2.$$

Proof. $\|x_t x_t^\top - S\|_F^2 = (x_t^\top x_t)^2 - 2x_t^\top S x_t + \|S\|_F^2$. Moreover $\sum_t x_t^\top S x_t = \text{tr}(S \sum_t x_t x_t^\top) = T \text{tr}(S^2) = T \|S\|_F^2$. Summing: $\sum_t (x_t^\top x_t)^2 - 2T \|S\|_F^2 + T \|S\|_F^2$. ■

The $\bar{b}^2$ term therefore costs $O(TN)$ on top of computing S , without forming the T matrices $x_t x_t^\top$.

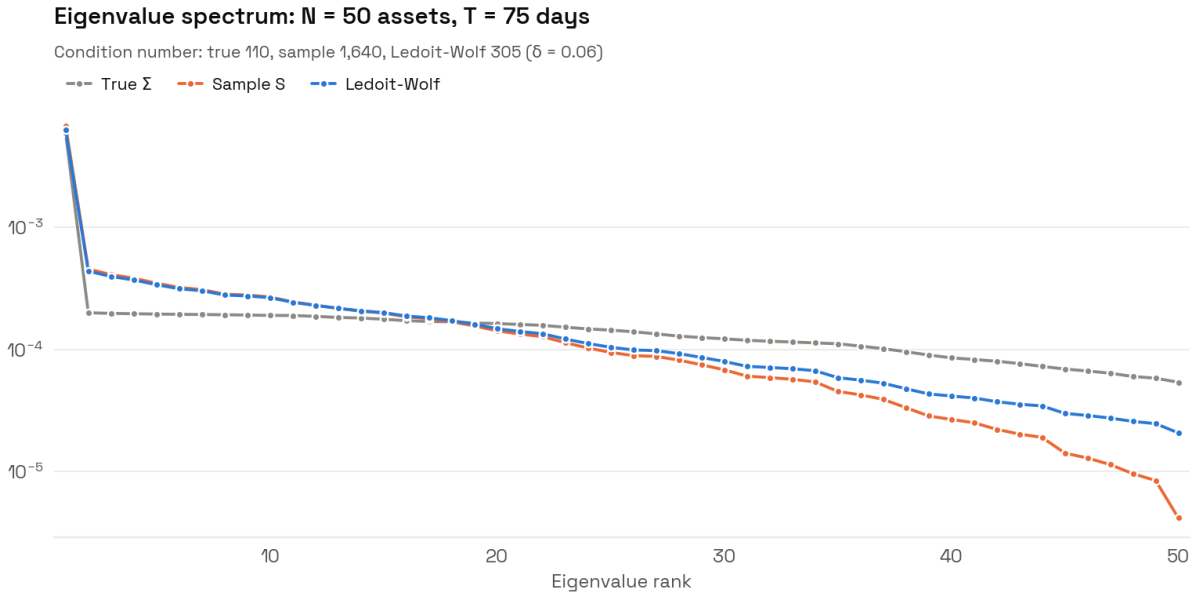


Figure 1. Eigenvalue spectrum, 50 assets and 75 days, true one-factor model. The sample covariance crushes the small eigenvalues (the last one is more than 10 times too small); Ledoit-Wolf lifts them and divides the condition number by 5.

2.3 EWMA

$$\hat{\Sigma}_t = \sum_{s=0}^{T-1} \omega_s (R_{t-s} - \bar{R})(R_{t-s} - \bar{R})^\top, \quad \omega_s \propto \lambda^s, \quad \lambda = 2^{-1/h},$$

with h the half-life (63 days by default), in the spirit of RiskMetrics (J.P. Morgan and Reuters, 1996). The effective number of observations $(\sum \omega_s)^2 / \sum \omega_s^2 \approx (1 + \lambda)/(1 - \lambda)$ is about 180 days for $h = 63$: more responsive to regime changes, but noisier.

Practitioner's tip

- The choice of covariance estimator often matters more than the choice between ERC and HRP.
- Means are much harder to estimate than covariances. The standard error of an annualized mean is $\sigma/\sqrt{Y}$ with Y the number of years: with $\sigma = 16\%$ and 10 years, $\pm 5\%$ on a mean of 8%. It does not depend on the sampling frequency. The error on a variance shrinks with the

number of observations: moving to daily data helps. This is the fundamental reason why every method in this document ignores expected returns.

- Beyond 50 assets, prefer a factor model $\Sigma = B\Sigma_F B^\top + D$ or nonlinear shrinkage (Ledoit and Wolf, 2020).

In the code

portopt/estimator.py: sample_cov, ledoit_wolf_cov, ewma_cov, estimate_cov. Every estimator accepts cov_method="sample" | "ledoit_wolf" | "ewma" or a callable returns -> ndarray.

3 The common building block: risk contributions

3.1 Euler decomposition

Proposition 3.1. *The volatility $\sigma_p(w) = \sqrt{w^\top \Sigma w}$ is homogeneous of degree 1 and*

$$\sigma_p(w) = \sum_{i=1}^N w_i \frac{\partial \sigma_p(w)}{\partial w_i}, \quad \frac{\partial \sigma_p}{\partial w} = \frac{\Sigma w}{\sigma_p}.$$

Proof. $\sigma_p(\lambda w) = \lambda \sigma_p(w)$ for $\lambda > 0$. Differentiating with respect to λ and evaluating at $\lambda = 1$ gives $\sum_i w_i \partial_i \sigma_p(w) = \sigma_p(w)$ (Euler's theorem for homogeneous functions). The gradient follows from differentiating $\sqrt{w^\top \Sigma w}$. ■

Definition 3.2. Marginal, total and relative risk contribution of asset i :

$$\text{MRC}_i = \frac{(\Sigma w)_i}{\sigma_p}, \quad \text{RC}_i = w_i \frac{(\Sigma w)_i}{\sigma_p}, \quad \frac{\text{RC}_i}{\sigma_p} = \frac{w_i (\Sigma w)_i}{w^\top \Sigma w}.$$

By proposition 3.1, $\sum_i \text{RC}_i = \sigma_p$ and the relative contributions sum to 1.

3.2 Interpretation

Since $(\Sigma w)_i = \text{Cov}(R_i, R_p)$,

$$\text{MRC}_i = \frac{\text{Cov}(R_i, R_p)}{\sigma_p} = \rho_{i,p} \sigma_i = \beta_{i,p} \sigma_p.$$

An asset contributes to risk in proportion to its **beta to the portfolio**, not to its own volatility. A very volatile asset that is uncorrelated with the portfolio contributes little; an asset with $\rho_{i,p} < 0$ has a negative contribution: it reduces risk.

Worked example

A 60/40 equity/bond portfolio, $\sigma_E = 16\%$, $\sigma_B = 6\%$, $\rho = -0.2$.

$$\begin{aligned} \sigma_p^2 &= 0.6^2 \cdot 0.16^2 + 0.4^2 \cdot 0.06^2 + 2 \cdot 0.6 \cdot 0.4 \cdot (-0.2) \cdot 0.16 \cdot 0.06 \\ &= 0.009216 + 0.000576 - 0.000922 = 0.008870, \quad \sigma_p = 9.4\%, \\ (\Sigma w)_E &= 0.6 \cdot 0.0256 + 0.4 \cdot (-0.00192) = 0.014592, \\ \text{RC}_E / \sigma_p &= 0.6 \cdot 0.014592 / 0.008870 = 98.7\%, \quad \text{RC}_B / \sigma_p = 1.3\%. \end{aligned}$$

A 60/40 portfolio in capital is a 99/1 portfolio in risk. This is the historical motivation for risk parity.

In the code

```
portopt.estimator.risk_contributions(w, cov); Backtester.risk_contributions()
returns the ex ante contributions at each rebalancing.
```

4 Allocation methods

All estimators share the same interface:

```
est = ERC(cov_method="ledoit_wolf").fit(returns_is) # T x N DataFrame, no NaN
est.weights_ # pd.Series, sums to 1, long-only
est.cov_ # covariance used
```

4.1 Equal weight (1/N)

$$w_i = \frac{1}{N}.$$

Intuition. Estimate nothing, so estimate nothing wrong.

Theory. If the N assets share the same volatility σ and the same correlation ρ ,

$$\sigma_p^2 = \frac{\sigma^2}{N} (1 + (N-1)\rho) \xrightarrow{N \rightarrow \infty} \rho \sigma^2 :$$

diversification removes idiosyncratic risk, not common risk.

Key result. DeMiguel et al. (2009) compare 14 optimization models (Markowitz, Bayes-Stein, constrained minimum variance, etc.) on 7 datasets: none significantly beats 1/N in out-of-sample Sharpe ratio. The estimation window needed for Markowitz to beat 1/N is around 3,000 months for 25 assets. 1/N is the **minimum benchmark** for any allocation research.

Limitations. Allocation in capital, not in risk: the most volatile asset dominates (section 3). The portfolio depends on how the universe is defined: adding 5 equity ETFs to a universe of 10 changes it radically.

Practitioner's tip

The turnover of 1/N is not zero: weights drift with prices and the portfolio must be rebalanced (38% a year in the example of section 9). A buy-and-hold 1/N is no longer 1/N after six months.

In the code

```
EqualWeight()
```

4.2 Inverse volatility

$$w_i = \frac{\sigma_i^{-k}}{\sum_j \sigma_j^{-k}}, \quad k = 1 \text{ (inverse volatility)}, \quad k = 2 \text{ (inverse variance)}.$$

Intuition. Give every asset the same volatility budget while ignoring correlations: $w_i \sigma_i$ is constant. Also called *naive risk parity*.

Proposition 4.1. (i) If all correlations are equal ($\rho_{ij} = \rho$ for $i \neq j$), the inverse volatility portfolio is the ERC portfolio.

(ii) If Σ is diagonal, the inverse variance portfolio is the minimum variance portfolio.

Proof. (i) With $w_i = c/\sigma_i$: $(\Sigma w)_i = \sum_j \rho_{ij} \sigma_i \sigma_j \frac{c}{\sigma_j} = \sigma_i c (1 + (N-1)\rho)$, so $w_i(\Sigma w)_i = c^2(1 + (N-1)\rho)$ does not depend on i . (ii) The minimum variance portfolio is $w \propto \Sigma^{-1} \mathbf{1}$ and, for a diagonal Σ , $\Sigma^{-1} \mathbf{1} = (1/\sigma_i^2)_i$. ■

InvVol is thus ERC under the assumption of homogeneous correlations, and inverse variance is minimum variance under the assumption of zero correlations. The gap between InvVol and ERC measures **the information carried by the correlation structure**.

Practitioner's tip

With two assets, InvVol and ERC coincide whatever ρ : there is only one correlation, so it is homogeneous.

In the code

```
InverseVolatility(exponent=1)
```

4.3 Risk budgeting

Problem. Given budgets $b_i > 0$ with $\sum_i b_i = 1$, find $w \geq 0$, $\mathbf{1}^\top w = 1$, such that

$$\frac{w_i(\Sigma w)_i}{w^\top \Sigma w} = b_i \quad \forall i.$$

This is a nonlinear system; the trick is to recast it as a convex problem.

Theorem 4.2 (Spinu, 2013; Roncalli, 2013). *Let $\Sigma \succ 0$ and*

$$y^* = \arg \min_{y \in \mathbb{R}_{++}^N} f(y), \quad f(y) = \frac{1}{2} y^\top \Sigma y - \sum_{i=1}^N b_i \ln y_i.$$

Then y^ exists and is unique, and $w^* = y^*/\mathbf{1}^\top y^*$ is the unique long-only risk budgeting portfolio. Moreover $y^{*\top} \Sigma y^* = 1$.*

Proof. *Existence.* f is continuous on $\mathbb{R}_{++}^N$. When some $y_i \rightarrow 0^+$, $-b_i \ln y_i \rightarrow +\infty$; when $\|y\| \rightarrow \infty$, $\frac{1}{2} y^\top \Sigma y \geq \frac{1}{2} \lambda_{\min} \|y\|^2$ dominates the logarithmic terms. f is therefore coercive and attains its minimum in the interior of $\mathbb{R}_{++}^N$.

Uniqueness. $\nabla^2 f(y) = \Sigma + \text{diag}(b_i/y_i^2) \succ 0$: f is strictly convex.

First-order condition. $\partial_i f(y) = (\Sigma y)_i - b_i/y_i = 0$, i.e. $y_i(\Sigma y)_i = b_i$: the unnormalized risk contributions of y^* are exactly the budgets.

Normalization. $y_i(\Sigma y)_i$ is homogeneous of degree 2, so the relative contributions of $w^* = y^*/\mathbf{1}^\top y^*$ equal $b_i/\sum_j b_j = b_i$.

Uniqueness of the portfolio. If w is a risk budgeting portfolio, then $y = w/\sqrt{w^\top \Sigma w}$ satisfies $y_i(\Sigma y)_i = b_i$, hence $\nabla f(y) = 0$, hence $y = y^*$ by strict convexity.

Finally, summing the first-order conditions gives $y^{*\top} \Sigma y^* = \sum_i b_i = 1$. ■

Algorithm: cyclical coordinate descent

We minimize f in y_i alone, the other coordinates held fixed (Griveau-Billion et al., 2013). The first-order condition in y_i reads

$$\Sigma_{ii} y_i^2 + c_i y_i - b_i = 0, \quad c_i = \sum_{j \neq i} \Sigma_{ij} y_j.$$

The discriminant $c_i^2 + 4\Sigma_{ii}b_i$ is positive and the product of the roots $-b_i/\Sigma_{ii}$ is negative: there is **exactly one positive root**, which gives the closed-form update

$$y_i \leftarrow \frac{-c_i + \sqrt{c_i^2 + 4\Sigma_{ii}b_i}}{2\Sigma_{ii}}.$$

Coordinate descent with exact minimization converges to the global optimum for a strictly convex function made of a smooth term plus a separable term (Tseng, 2001).

Algorithm 1: Risk budgeting by coordinate descent (`RiskParity._ccd`)

Input: $\Sigma \succ 0$, budgets b , tolerance $\varepsilon = 10^{-10}$

Output: weights w

```

1  $y_i \leftarrow b_i/\sigma_i$  for all  $i$ ;    $y \leftarrow y/\sqrt{y^\top \Sigma y}$ ;
2  $s \leftarrow \Sigma y$ ;                                     # kept up to date, never recomputed
3 repeat
4    $y^{\text{old}} \leftarrow y$ ;
5   for  $i \leftarrow 1$  to  $N$  do
6      $c \leftarrow s_i - \Sigma_{ii}y_i$ ;
7      $y'_i \leftarrow (-c + \sqrt{c^2 + 4\Sigma_{ii}b_i})/(2\Sigma_{ii})$ ;
8      $s \leftarrow s + \Sigma_{:,i}(y'_i - y_i)$ ;    $y_i \leftarrow y'_i$ ;           #  $O(N)$ 
9 until  $\|y - y^{\text{old}}\| \leq \varepsilon \|y\|$ ;
10 return  $w = y/\mathbf{1}^\top y$ 
```

The vector Σy is updated incrementally: $O(N)$ per coordinate, $O(N^2)$ per sweep, and **no matrix inversion**. The initialization $y \propto b_i/\sigma_i$, rescaled to $y^\top \Sigma y = 1$, starts close to the solution; in practice 10 to 50 sweeps are enough.

Using budgets. Express a view in risk rather than in capital. In the example, the “RB bond tilt” strategy doubles the budget of TLT, IEF and LQD: the three carry 45% of the risk (15% each) versus 30% under ERC.

Practitioner’s tip

- Budgets must be strictly positive (log barrier). To exclude an asset, remove it from the universe.
- Without the long-only constraint, the system $y_i(\Sigma y)_i = b_i$ has one solution per orthant, i.e. up to 2^{N-1} distinct portfolios (Bai et al., 2016). The long-only solution is the only one you want in practice.
- To target a volatility (say 10%), scale w by $0.10/\sigma_p$: relative contributions are invariant to leverage.

In the code

`RiskParity(budgets={...})`, method `_ccd`, attribute `n_iter_` (number of sweeps).

4.4 Equal Risk Contribution (ERC)

ERC is risk budgeting with $b_i = 1/N$: every asset carries the same share of risk.

Theorem 4.3 (Maillard et al., 2010). *Let σ_{MV} be the volatility of the long-only minimum variance portfolio. Then*

$$\sigma_{\text{MV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}.$$

Proof. The first inequality is immediate: the long-only minimum variance portfolio minimizes σ_p over the whole simplex, which contains w_{ERC} .

For the second, $\sigma_p(w) = \|\Sigma^{1/2}w\|$ is convex, hence lies above its tangent at w_{ERC} :

$$\sigma_p\left(\frac{1}{N}\right) \geq \sigma_{\text{ERC}} + \nabla \sigma_p(w_{\text{ERC}})^\top \left(\frac{1}{N} - w_{\text{ERC}}\right) = \frac{1}{N} \sum_i \partial_i \sigma_p(w_{\text{ERC}}),$$

since $\nabla \sigma_p(w)^\top w = \sigma_p(w)$ (proposition 3.1). At ERC, $w_i \partial_i \sigma_p = \sigma_{\text{ERC}}/N$, so $\partial_i \sigma_p = \sigma_{\text{ERC}}/(Nw_i)$ and

$$\sigma_{1/N} \geq \frac{\sigma_{\text{ERC}}}{N^2} \sum_i \frac{1}{w_i} \geq \sigma_{\text{ERC}},$$

by the arithmetic-harmonic mean inequality: $\sum_i 1/w_i \geq N^2 / \sum_i w_i = N^2$. ■

Interpretation. In the parametrization of theorem 4.2, y^* also minimizes $\sqrt{y^\top \Sigma y}$ subject to $\sum_i \ln y_i \geq c$ (the value of c only changes the scale of y^*). ERC is therefore a minimum variance portfolio under a logarithmic **diversification constraint**: without the constraint we recover minimum variance; the constraint prevents any weight from going to zero.

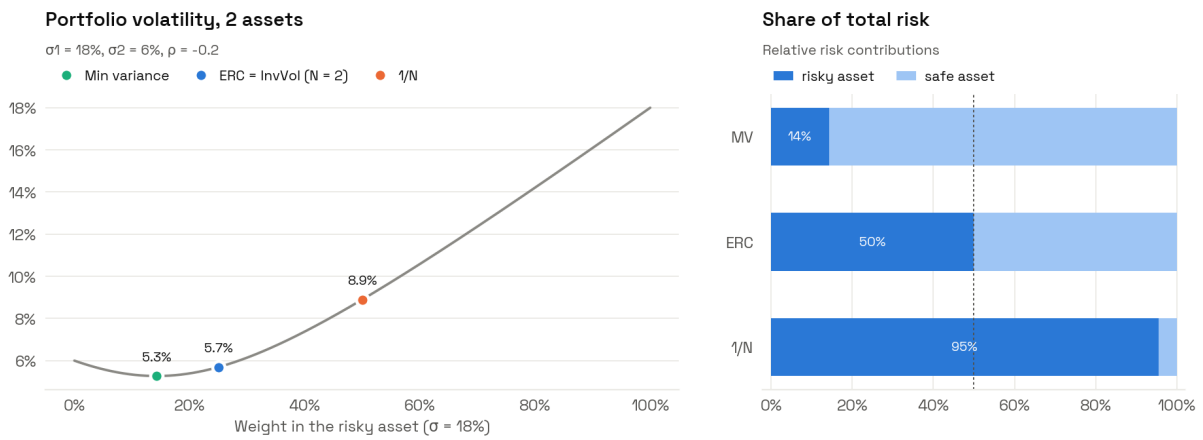


Figure 2. Two assets, $\sigma_1 = 18\%$, $\sigma_2 = 6\%$, $\rho = -0.2$. Minimum variance (5.3%) puts 14% of the risk on the risky asset, $1/N$ (8.9%) puts 95%, ERC (5.7%) exactly 50%. ERC's volatility is barely above the minimum: a lot of diversification is bought for very little variance.

Proposition 4.4. *If all assets have the same Sharpe ratio and correlations are homogeneous, ERC is the tangency (maximum Sharpe) portfolio.*

Proof. The tangency portfolio is $w \propto \Sigma^{-1}\mu$. With $\mu_i = s\sigma_i$ and $\Sigma = DCD$, $D = \text{diag}(\sigma_i)$, $C = (1-\rho)I + \rho\mathbf{1}\mathbf{1}^\top$: $\Sigma^{-1}\mu = sD^{-1}C^{-1}\mathbf{1}$. Since $C\mathbf{1} = (1+(N-1)\rho)\mathbf{1}$, $C^{-1}\mathbf{1} = \mathbf{1}/(1+(N-1)\rho)$ and $w \propto D^{-1}\mathbf{1} = (1/\sigma_i)_i$, which is ERC by proposition 4.1. ■

Why ERC works out of sample. It does not use expected returns, the most poorly estimated quantity. It does not invert Σ : weights depend continuously and stably on Σ , whereas minimum variance amplifies the noise in small eigenvalues. And proposition 4.4 makes it optimal under a reasonable “ignorance” assumption.

Practitioner's tip

- ERC overweights low-volatility assets (bonds). Its historical performance owes a lot to the 1981–2020 bond bull market. In 2022, stocks and bonds fell together and levered risk parity funds lost around 20% or more.
- An unlevered ERC has low volatility (8% in the example). Compare it with $1/N$ on Sharpe ratio, or at equal volatility.

In the code

```
ERC(cov_method=...)
```

4.5 Hierarchical Risk Parity (HRP)

Intuition (López de Prado, 2016). Markowitz treats every asset as a potential substitute for every other (complete graph). But assets have a hierarchical structure: SPY and IWM are close, so are TLT and IEF, and these two families are far apart. HRP allocates between families first, then within them.

Step 1: distance

$$d_{ij} = \sqrt{\frac{1}{2}(1 - \rho_{ij})} \in [0, 1].$$

Proposition 4.5. d is a metric.

Proof. Let z_i be the series of asset i , demeaned and scaled to unit norm. Then $\rho_{ij} = z_i^\top z_j$ and $\|z_i - z_j\|^2 = 2(1 - \rho_{ij})$, so $d_{ij} = \frac{1}{2}\|z_i - z_j\|$. Non-negativity, symmetry and the triangle inequality are inherited from the Euclidean norm; $d_{ij} = 0$ if and only if $\rho_{ij} = 1$. ■

Step 2: hierarchical clustering

Repeatedly merge the two closest clusters. The distance between clusters depends on the linkage: **single** (minimum distance, original paper, prone to chaining), **complete** (maximum, compact clusters), **average** (mean, a good compromise), **ward** (minimizes the increase in within-cluster variance).

Step 3: quasi-diagonalization

Reorder the assets following the leaves of the dendrogram. Correlated assets become neighbours and the reordered covariance matrix concentrates its large entries near the diagonal.

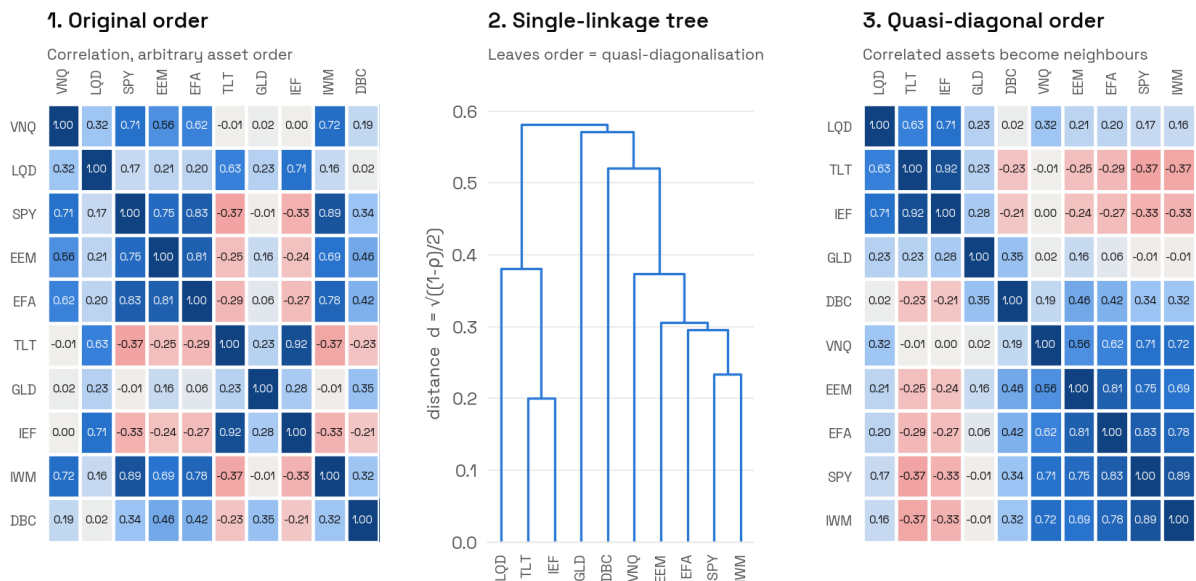


Figure 3. HRP steps. In the arbitrary order (left), the structure is invisible. After quasi-diagonalization (right), two blocks appear: rates and credit (LQD, TLT, IEF) at the top, risky assets (VNQ, EEM, EFA, SPY, IWM) at the bottom, with gold and commodities in between.

Step 4: recursive bisection

Start from $w = \mathbf{1}$. Each time a cluster is split into two sub-clusters L and R :

$$V_c = \tilde{w}_c^\top \Sigma_c \tilde{w}_c, \quad \tilde{w}_c = \frac{\text{diag}(\Sigma_c)^{-1}}{\mathbf{1}^\top \text{diag}(\Sigma_c)^{-1}}, \quad \alpha = 1 - \frac{V_L}{V_L + V_R}, \quad w_L \leftarrow \alpha w_L, \quad w_R \leftarrow (1 - \alpha) w_R.$$

V_c is the variance of the inverse variance portfolio of cluster c : each sub-cluster receives a share inversely proportional to its variance. It is an inverse variance allocation between two super-assets.

Algorithm 2: Hierarchical Risk Parity (HRP)

Input: in-sample returns, `linkage`, `split`

Output: weights w

```

1  $\Sigma \leftarrow$  estimated covariance;  $\rho \leftarrow$  correlation;
2  $d_{ij} \leftarrow \sqrt{(1 - \rho_{ij})/2}$ ;
3 tree  $\leftarrow$  hierarchical clustering of  $d$  (condensed form);
4 order  $\leftarrow$  leaves of the tree ;                                # quasi-diagonalization
5  $w \leftarrow \mathbf{1}$ ; stack  $\leftarrow$  [order];
6 while stack not empty do
7   pop a cluster  $c$  from the stack;
8   if  $|c| = 1$  then continue;
9    $(L, R) \leftarrow$  if split="bisection" then
10    | two halves of the list  $c$ 
11   else
12    | two children of node  $c$ 
13    $\alpha \leftarrow 1 - V_L/(V_L + V_R)$ ;
14    $w_L \leftarrow \alpha w_L$ ;  $w_R \leftarrow (1 - \alpha) w_R$ ;
15   push  $L$  and  $R$  onto the stack;
16 return  $w$ 

```

Proposition 4.6. (i) No matrix inversion is needed: HRP works even if Σ is singular ($T < N$).
(ii) If $N = 2$ or if Σ is diagonal, HRP is the inverse variance portfolio.

Proof. (i) Only the diagonals of sub-blocks are inverted. (ii) For a diagonal Σ , $V_c = 1/S_c$ with $S_c = \sum_{i \in c} \sigma_i^{-2}$, so $\alpha = V_R/(V_L + V_R) = S_L/(S_L + S_R)$: cluster L receives the inverse variance share of its assets. By induction on the tree, each asset receives $\sigma_i^{-2} / \sum_j \sigma_j^{-2}$. The case $N = 2$ is a single split between two singletons, with $V_c = \sigma_c^2$. ■

Two splitting variants (`split`=). "bisection" (original paper) cuts the *ordered list* into two equal halves, ignoring the tree: a natural cluster of 3 assets in a universe of 10 can be cut in two. "dendrogram" follows the actual nodes of the dendrogram; it is the recommended variant.

Practitioner's tip

- López de Prado's original code passes the *square* matrix d to `scipy.cluster.hierarchy.linkage`. SciPy then treats each row as an observation and computes $\tilde{d}_{ij} = \|d_{\cdot, i} - d_{\cdot, j}\|_2$, a distance of distances. Many implementations reproduce this behaviour without knowing it. `portopt` passes the condensed form (`squareform`), hence d itself.
- HRP is unstable over time: a small change in correlation can reorganize the tree and move a lot of weight. In the example its turnover (120% a year) is 2.5 times that of ERC; at 20 bps of costs, that is 24 bps of performance a year. Try `linkage="ward"` or `"average"`.
- HRP concentrates risk in the least volatile cluster (71% of the risk on TLT, IEF and LQD in

the example). It is a defensive portfolio more than a risk-diversified one.

In the code

```
HRP(linkage="single", split="bisection"); attributes order_ (quasi-diagonal order) and
linkage_matrix_ (to pass to scipy.cluster.hierarchy.dendrogram).
```

4.6 Summary

	1/N	InvVol	Risk budgeting / ERC	HRP
Uses μ	no	no	no	no
Uses σ_i	no	yes	yes	yes
Uses ρ_{ij}	no	no	yes	yes (structure)
Inverts Σ	no	no	no	no
Solution	direct	direct	iterative, unique	direct, tree-dependent
Complexity	$O(N)$	$O(N)$	$O(N^2)$ per sweep	$O(N^2)$
Equalizes	capital	$w_i \sigma_i$	$w_i (\Sigma w)_i$	variance across clusters
Weight stability	perfect	very good	good	average
Weak spot	risk concentration	ignores correlations	needs leverage	turnover, linkage sensitivity

Table 1. Comparison of allocation methods.

5 Walk-forward backtest

Walk-forward backtest (rolling window)

Weights fitted on returns up to $t-1$ are held on $[t, t+H)$, then the window rolls by H

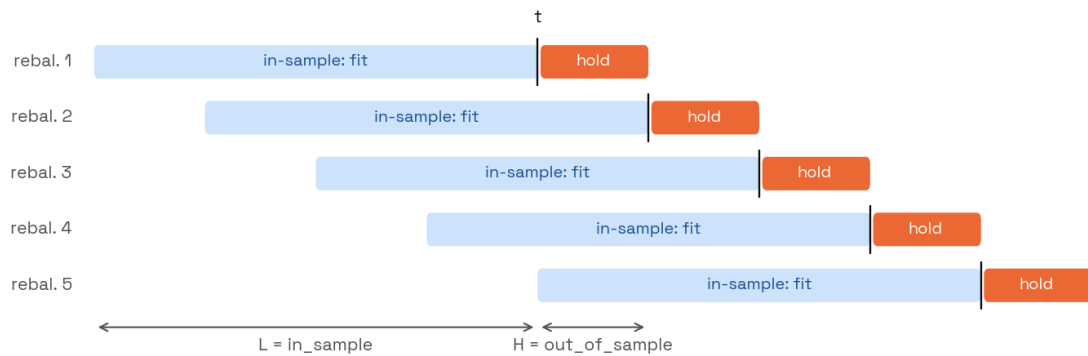


Figure 4. Rolling-window walk-forward backtest. Weights estimated on returns up to $t - 1$ are held over $[t, t + H)$, then the window moves forward by H .

5.1 Mechanics

Algorithm 3: Walk-forward backtest (`Backtester.run`)

Input: returns $R_{0..T-1}$, estimator, L , H , cost c
Output: out-of-sample returns net of costs
 1 $w^{\text{drift}} \leftarrow 0$; # start in cash
 2 $t \leftarrow L$;
 3 **while** $t < T$ **do**
 4 $\text{window} \leftarrow R_{t-L..t-1}$ (**rolling**) or $R_{0..t-1}$ (**expanding**);
 5 $\text{universe} \leftarrow$ assets with no NaN over the window;
 6 $w \leftarrow$ copy of the estimator fitted on the window;
 7 $\tau \leftarrow \sum_i |w_i - w_i^{\text{drift}}|$;
 8 hold w over $R_{t..t+H-1}$ with weight drift; wealth $\times (1 - c\tau)$ at t ;
 9 $w^{\text{drift}} \leftarrow$ drifted weights at period end; $t \leftarrow t + H$;

Weights use information up to $t - 1$ and earn R_t : no look-ahead bias. A unit test checks this with a spy estimator that records the last date it sees. The estimator is copied at every date: no state leaks from one window to the next.

5.2 Weight drift

Between two rebalancings, a real portfolio does not keep constant weights: they drift with prices. For $k = 1, \dots, H$,

$$V_k = \sum_i w_i \prod_{s=1}^k (1 + R_{i,s}), \quad R_{p,k} = \frac{V_k}{V_{k-1}} - 1, \quad w_i^{\text{drift}} = \frac{w_i \prod_{s=1}^H (1 + R_{i,s})}{V_H}.$$

The implementation is vectorized: `np.cumprod(1 + R, axis=0) @ w` gives the whole path $V_1, \dots, V_H$ at once. With `drift=False`, the portfolio is assumed to be rebalanced to target every day, for free: more optimistic than reality.

5.3 Transaction costs

At each rebalancing, one-way turnover is measured against the **drifted weights**:

$$\tau_t = \sum_i |w_i^{\text{target}} - w_i^{\text{drift}}|.$$

Measuring against the previous target is a common mistake: if equities rose 10% during the month, an ERC portfolio must sell some even if its target did not move. Ignoring drift understates turnover.

Wealth is multiplied by $(1 - c\tau_t)$ at rebalancing, with the cost c given in basis points through `transaction_cost_bps`. The first rebalancing starts from cash ($\tau = 1$). Annualized turnover, excluding the initial build-up, is

$$\text{Turnover}_{\text{ann}} = \frac{\sum_{t>t_0} \tau_t}{\text{number of years}},$$

and the annual cost is about $c \times \text{Turnover}_{\text{ann}}$: 120% turnover at 10 bps costs 12 bps a year.

Practitioner's tip

- (L, H) is a **hyperparameter**. Test a grid and look at the stability of the ranking of methods, never at the best pair: otherwise you are overfitting the backtest.

- **Timing luck.** Shifting the first rebalancing date by 0 to $H - 1$ days gives H different paths for the same strategy. Their dispersion measures how much of the result is luck; for a monthly strategy it is often of the same order as the gaps between methods.
- `window="expanding"` reduces estimation noise but reacts slowly to regimes. A rolling window of 252 to 756 days is the standard.

In the code

`Backtester(returns, estimator, in_sample, out_of_sample, window, transaction_cost_bps, drift); .run()` returns the net out-of-sample returns; attributes `weights_`, `turnover_`, `gross_returns_`; methods `annualized_turnover()` and `risk_contributions()`. `run_backtests(returns, {name: est}, ...)` compares several estimators on the same grid.

6 Performance metrics

Every metric takes a `pd.Series` (one strategy) or a `pd.DataFrame` (one column per strategy) of periodic simple returns. `summary()` produces the full table.

6.1 Total return and CAGR

$$\text{TR} = \prod_{t=1}^T (1 + R_t) - 1, \quad \text{CAGR} = (1 + \text{TR})^{P/T} - 1.$$

The CAGR is the constant rate that yields the same final wealth: it is the return the investor actually experiences. The annualized arithmetic mean is $\mu_{\text{ann}} = P \bar{R}$.

6.2 Volatility drag

Proposition 6.1. *The log growth rate $g = \ln(1 + \text{CAGR})$ satisfies, to second order,*

$$g \approx \mu_{\text{ann}} - \frac{\sigma_{\text{ann}}^2}{2}.$$

Proof. $g = \frac{P}{T} \sum_t \ln(1 + R_t)$. With $\ln(1 + R) = R - R^2/2 + O(R^3)$ and $\frac{1}{T} \sum_t R_t^2 = \hat{\sigma}^2 + \bar{R}^2 \approx \hat{\sigma}^2$ for daily data, $g \approx P \bar{R} - P \hat{\sigma}^2/2 = \mu_{\text{ann}} - \sigma_{\text{ann}}^2/2$. ■

Worked example

Two strategies with $\mu_{\text{ann}} = 8\%$: at $\sigma = 10\%$, $g \approx 7.5\%$; at $\sigma = 30\%$, $g \approx 3.5\%$. In the example of section 9, 1/N has $\mu_{\text{ann}} = 7.36\%$ and $\sigma_{\text{ann}} = 11.42\%$: the formula predicts $g = 7.36\% - 0.65\% = 6.71\%$, the realized value is $\ln(1.0693) = 6.70\%$.

Practitioner's tip

This is the strongest argument for variance reduction: at equal mean, lower volatility means higher final wealth. The multi-asset version is the *rebalancing premium* or *excess growth rate* (Fernholz and Shay, 1982; Booth and Fama, 1992): for a constant-weight portfolio, $g_p - \sum_i w_i g_i \approx \frac{1}{2} (\sum_i w_i \sigma_i^2 - \sigma_p^2) \geq 0$.

6.3 Sharpe ratio

$$\text{SR} = \sqrt{P} \frac{\overline{R - r_f}}{\hat{\sigma}}, \quad r_{f,\text{period}} = (1 + r_{f,\text{ann}})^{1/P} - 1.$$

Excess return per unit of total risk. It is **invariant to leverage**: leveraging twice doubles both the excess return and the volatility. A mean-variance investor picks the highest-Sharpe strategy and then sets the leverage.

Annualization. The $\sqrt{P}$ factor is only valid for iid returns. For AR(1) returns with autocorrelation ρ_1 and a long horizon of q periods (Lo, 2002),

$$SR_q \approx SR_1 \cdot \sqrt{q} \cdot \sqrt{\frac{1 - \rho_1}{1 + \rho_1}}.$$

Smoothed returns ($\rho_1 > 0$: illiquid hedge funds, private equity) have an overstated $\sqrt{P}$ -annualized Sharpe ratio.

6.4 Sortino ratio

$$\text{Sortino} = \sqrt{P} \frac{\overline{R - \tau}}{DD_\tau}, \quad DD_\tau = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t - \tau, 0)^2}.$$

Only downside volatility is penalized. The sum in the denominator runs over *all* observations (positive ones count as zero): dividing by the number of negative observations is a common mistake that inflates the downside deviation. For a symmetric distribution with a mean small relative to σ , $DD_0 \approx \sigma/\sqrt{2}$, so Sortino $\approx \sqrt{2}$ SR; a ratio well above 1.41 signals positive skewness.

6.5 Calmar ratio

$$\text{Calmar} = \frac{\text{CAGR}}{|\text{MaxDD}|}.$$

The number of years of return needed to make up for the worst loss. It depends heavily on the length of the history (MaxDD can only get worse as T grows): only compare strategies over the same period.

6.6 Hit ratio

$$\text{Hit} = \frac{1}{T} \sum_t \mathbb{1}\{R_t > 0\}.$$

Close to 50–55% for daily allocation strategies, and not very informative on its own: a 40% hit ratio with gains twice as large as losses is excellent (trend following), a 90% hit ratio can hide a short volatility position (frequent small gains, rare and huge losses).

In the code

```
portopt/metric.py: cagr, annualized_mean, annualized_volatility, sharpe_ratio,
sortino_ratio, calmar_ratio, hit_ratio, summary.
```

7 Risk metrics

7.1 Annualized volatility

$\sigma_{\text{ann}} = \hat{\sigma}\sqrt{P}$. A symmetric measure: a 5% gain “costs” as much as a 5% loss. It is insufficient for skewed or fat-tailed distributions (section 7.4).

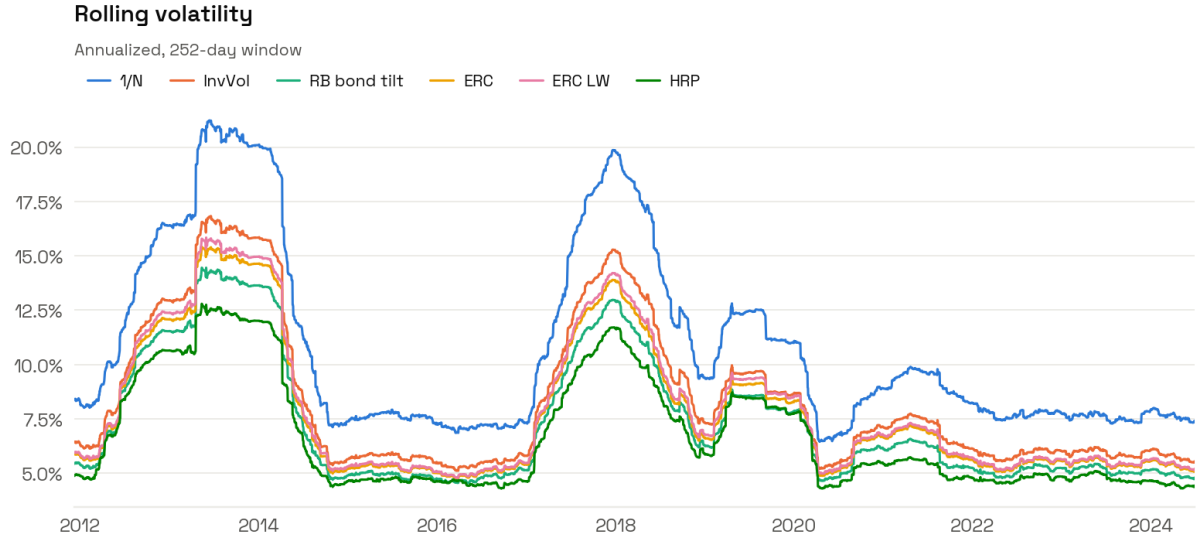


Figure 5. 252-day rolling volatility. Realized volatility clusters in regimes: the simulated stress episodes double the volatility of every strategy, but the gap between strategies stays stable.

7.2 Drawdown

Definition 7.1. With wealth $W_t = \prod_{s \leq t} (1 + R_s)$ and $W_0 = 1$,

$$DD_t = \frac{W_t}{\max(1, \max_{s \leq t} W_s)} - 1, \quad \text{MaxDD} = \min_t DD_t.$$

The maximum time under water is the longest consecutive stretch with $DD_t < 0$.

Initial capital counts as a peak: a strategy that loses 10% on day one has a drawdown of -10% , even if W never went above 0.9.



Figure 6. Anatomy of a drawdown: peak, trough, recovery and time under water (1/N portfolio, simulated data).

Asymmetry of losses. Recovering a loss ℓ requires a gain $\ell/(1 - \ell)$: -20% requires $+25\%$, -50% requires $+100\%$.

Practitioner's tip

For a Brownian motion with drift μ and volatility σ (Magdon-Ismail et al., 2004), $\mathbb{E}[\text{MaxDD}] = \sqrt{\pi/2} \sigma \sqrt{T}$ if $\mu = 0$, and $\mathbb{E}[\text{MaxDD}] \sim \frac{\sigma^2}{2\mu} \ln T$ if $\mu > 0$. MaxDD therefore depends mechanically on the length of the backtest: -15% over 5 years is not comparable to -15% over 20 years.

Drawdowns

Peak-to-trough loss of wealth, dot = maximum drawdown

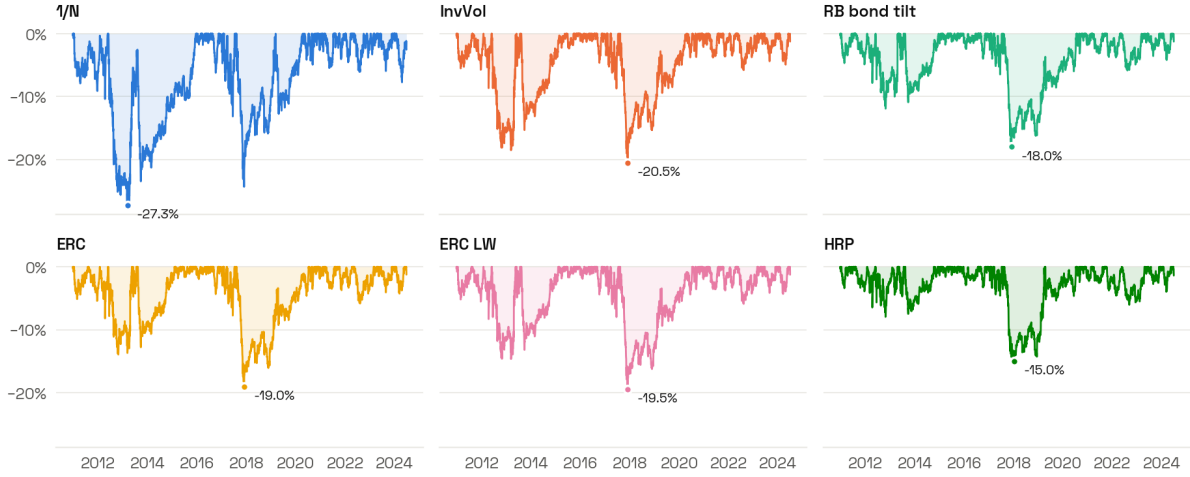


Figure 7. Drawdowns of the six strategies in the example; the dot marks the maximum drawdown.

7.3 Historical VaR and CVaR

Definition 7.2. With $q_\alpha(R)$ the empirical quantile of order α (5% by default),

$$\text{VaR}_\alpha = -q_\alpha(R), \quad \text{CVaR}_\alpha = -\mathbb{E}[R \mid R \leq q_\alpha(R)],$$

both expressed as a positive loss.

The 95% VaR says: “on 95% of days, we lose no more than this”, and says nothing about the other 5%. CVaR (expected shortfall) says: “on the worst 5% of days, we lose this on average”.

Proposition 7.3. *VaR is not subadditive; CVaR is a coherent risk measure (Artzner et al., 1999): monotone, translation invariant, positively homogeneous and subadditive.*

Counterexample to subadditivity

Two independent bonds, each with a 4% default probability (loss 100, otherwise 0). For each, $\text{VaR}_{95\%} = 0$ since default lies beyond the 5% quantile. For the portfolio of both, losses are 0, 100 and 200 with probabilities 92.16%, 7.68% and 0.16%: $\text{VaR}_{95\%} = 100 > 0 + 0$. VaR penalizes diversification. On the CVaR side, each bond has $\text{CVaR}_{95\%} = (4\% \times 100)/5\% = 80$, and the portfolio $(0.16\% \times 200 + 4.84\% \times 100)/5\% = 103.2 \leq 160$: subadditivity holds.

The FRTB framework (Basel III) replaced 99% VaR with 97.5% expected shortfall for market risk.

Practitioner's tip

Gaussian parametric VaR $-(\mu + z_\alpha \sigma)$ systematically underestimates tail risk. To stay parametric, use the Cornish-Fisher expansion, which corrects the quantile for skewness and kurtosis.

Daily return distribution and tail risk

1/N portfolio, excess kurtosis 14.1: the normal curve underestimates the tail (shaded)

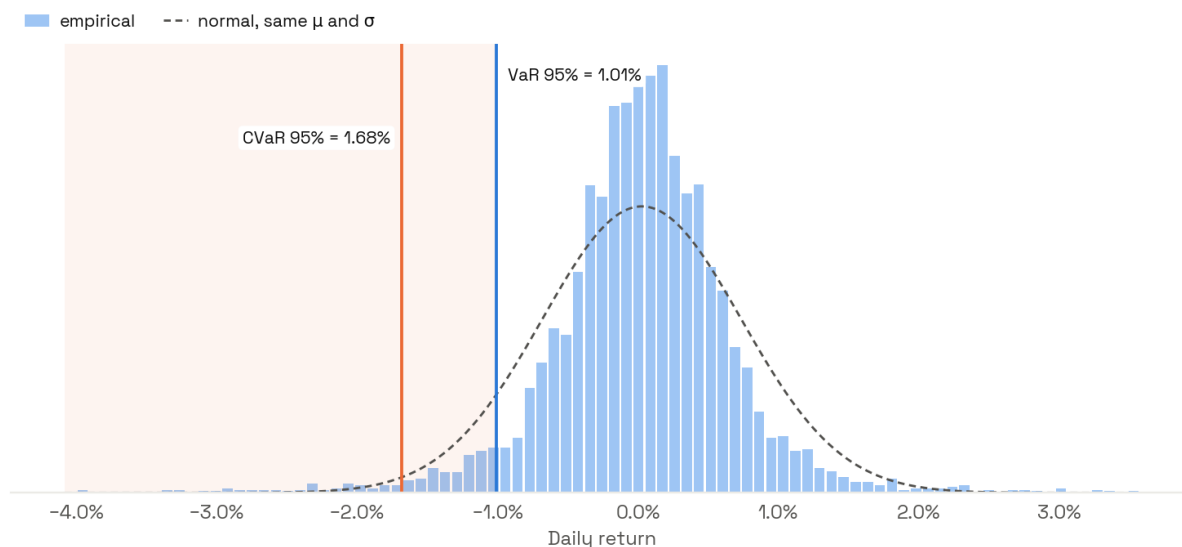


Figure 8. Distribution of daily returns of the 1/N portfolio. The normal distribution with the same mean and standard deviation underestimates the left tail (shaded area).

7.4 Skewness and kurtosis

$$\gamma_3 = \frac{\mathbb{E}[(R - \mu)^3]}{\sigma^3}, \quad \gamma_4 = \frac{\mathbb{E}[(R - \mu)^4]}{\sigma^4}, \quad \text{excess kurtosis} = \gamma_4 - 3.$$

$\gamma_3 < 0$ signals a fat left tail (rare but large losses: carry, option selling, credit). $\gamma_4 - 3 > 0$ (leptokurtic) signals more quiet days and more extreme days than a Gaussian; at daily frequency, 5 to 20 is common for equity indices.

Practitioner's tip

These higher moments are very poorly estimated: a single extreme day can flip skewness from -1 to $+1$. Look at them over subperiods before drawing any conclusion.

7.5 Turnover

Defined in section 5. It is an implementation risk metric: high turnover means exposure to costs, market impact and limited capacity.

In the code

```
max_drawdown, max_drawdown_duration, drawdown_series, value_at_risk,
conditional_value_at_risk, skewness, excess_kurtosis.
```

8 Statistical significance

A backtest is one realization of a random process. Before concluding that one strategy beats another, check that the gap can be told apart from noise.

8.1 Standard error of the Sharpe ratio

For iid Gaussian returns (Lo, 2002), in annual units over Y years,

$$\text{se}(\widehat{\text{SR}}) \approx \sqrt{\frac{1 + \frac{1}{2}\text{SR}_{\text{period}}^2}{Y}} \approx \frac{1}{\sqrt{Y}}.$$

How precise is an estimated Sharpe ratio?

95% confidence band, iid returns: $\text{se} \approx \sqrt{[(1 + \text{SR}^2/2) / T]}$, T in years

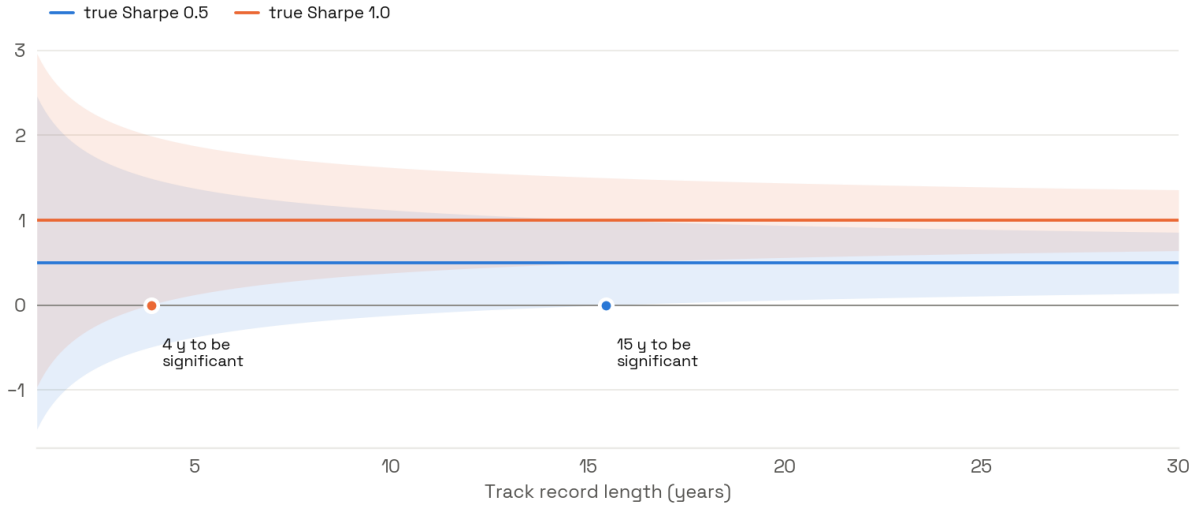


Figure 9. 95% confidence band of the estimated Sharpe ratio as a function of track record length.

For a Sharpe ratio of 0.5 to be significantly positive at 95%, one needs $Y \geq (1.96/0.5)^2 \approx 15$ years; for a Sharpe ratio of 1, about 4 years. Most 10-year backtests cannot tell a Sharpe ratio of 0.5 from zero.

Consequence for the example

Sharpe ratios range from 0.64 ($1/N$) to 0.88 (HRP) over 14 years, with a standard error of about $1/\sqrt{14} = 0.27$. The 0.24 gap is less than one standard error: we cannot claim that HRP beats $1/N$. The test on the Sharpe difference between two correlated strategies (Jobson and Korkie, 1981; Memmel, 2003) has a smaller standard error, but of the same order here.

8.2 Probabilistic Sharpe Ratio

Definition 8.1 (Bailey and López de Prado, 2012).

$$\text{PSR}(\text{SR}^*) = \Phi \left(\frac{(\widehat{\text{SR}} - \text{SR}^*)\sqrt{T-1}}{\sqrt{1 - \gamma_3 \widehat{\text{SR}} + \frac{\gamma_4 - 1}{4} \widehat{\text{SR}}^2}} \right),$$

where $\widehat{\text{SR}}$ and SR^* are expressed *per period* (not annualized), γ_3 is the skewness and γ_4 the (non-excess) kurtosis. It is the probability that the true Sharpe ratio exceeds the threshold SR^* .

Reading the denominator. Negative skewness or high kurtosis increase the variance of $\widehat{\text{SR}}$, and therefore lower the PSR. A short volatility strategy with a nice Sharpe ratio but $\gamma_3 = -3$ will have a much lower PSR than a trend following strategy with the same Sharpe ratio and $\gamma_3 > 0$. `summary()` reports $\text{PSR}(\text{SR} > 0)$; above 0.95 the Sharpe ratio is significantly positive.

8.3 Deflated Sharpe Ratio

If K variants (parameters, universes, methods) were tested and the best one kept, its Sharpe ratio is biased upward: even if all true Sharpe ratios are zero, the maximum of K estimated ones is positive. The DSR (Bailey and López de Prado, 2014) replaces SR^* in the PSR with the expected maximum of K zero Sharpe ratios:

$$SR_K^* \approx \sqrt{\text{Var}(\widehat{SR}_k)} \left[(1 - \gamma_E) \Phi^{-1}\left(1 - \frac{1}{K}\right) + \gamma_E \Phi^{-1}\left(1 - \frac{1}{Ke}\right) \right],$$

with $\gamma_E \approx 0.5772$ the Euler-Mascheroni constant and $\text{Var}(\widehat{SR}_k)$ the variance of the Sharpe ratios observed across the K trials. With $K = 100$, the bracket equals $0.4228 \times 2.326 + 0.5772 \times 2.680 \approx 2.53$: an observed Sharpe ratio of about 2.5 standard errors is needed to rule out pure data mining.

Practitioner's tip

Record the number of variants you tested. It is as important a piece of the backtest as the Sharpe ratio itself.

In the code

`probabilistic_sharpe_ratio(r, sr_benchmark=0.0)`; the `sr_benchmark` threshold is given annualized and converted internally.

9 Reading the example results

`python examples/run_backtest.py --synthetic`: 10 simulated ETFs, 252-day in-sample window, rebalancing every 21 days, 5 bps of costs, December 2010 to June 2024 (3,528 out-of-sample days).

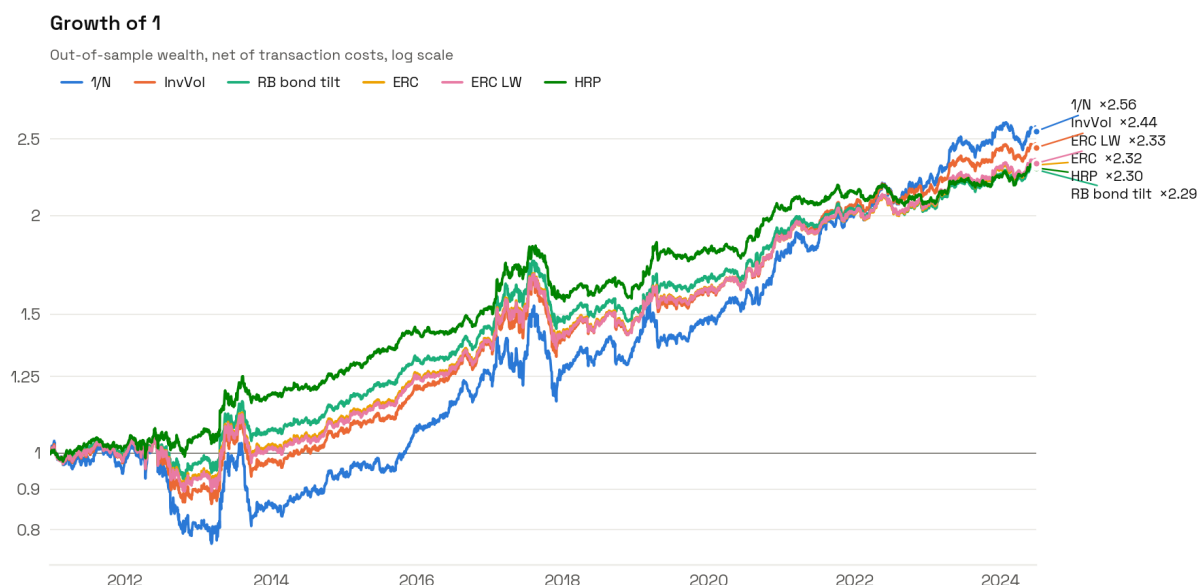


Figure 10. Out-of-sample growth of 1, net of costs, log scale.

Takeaways.

1. **1/N has the best CAGR but the worst risk.** It is simply more exposed to the risky assets, which had the best returns over the period. Its Sharpe ratio is the lowest.

	1/N	InvVol	RB bond tilt	ERC	ERC LW	HRP
CAGR	6.9%	6.6%	6.1%	6.2%	6.2%	6.1%
Volatility	11.4%	8.9%	7.7%	8.2%	8.4%	7.0%
Sharpe	0.64	0.76	0.81	0.77	0.76	0.88
Sortino	0.93	1.11	1.20	1.14	1.12	1.33
Calmar	0.25	0.32	0.34	0.33	0.32	0.41
Max DD	−27.3%	−20.5%	−18.0%	−19.0%	−19.5%	−15.0%
CVaR 95% (daily)	1.7%	1.3%	1.1%	1.2%	1.2%	1.0%
Annual turnover	38%	41%	45%	49%	48%	120%

Table 2. Out-of-sample metrics, net of costs.

Key metrics

Out-of-sample, net of costs. Turnover is one-way, per year

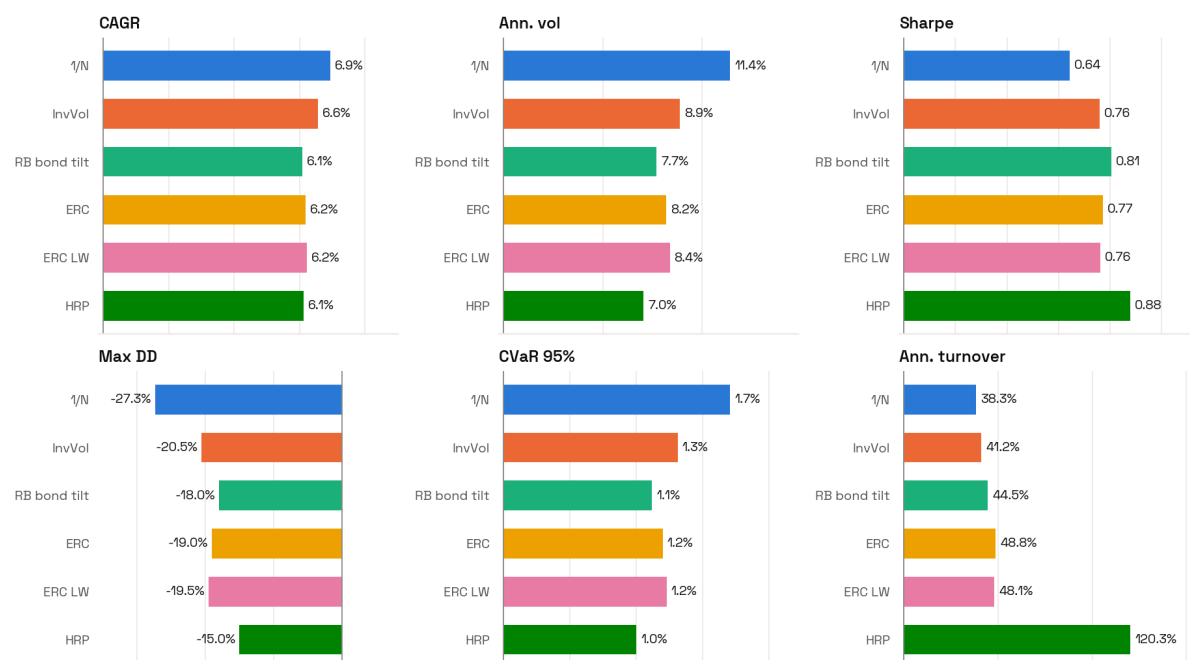


Figure 11. Key metrics by strategy.

2. **Every risk-based method improves the Sharpe ratio and reduces drawdown**, by 25 to 45%. This is the robust result from the literature.
3. **The ranking between ERC, InvVol and HRP is not significant** (section 8). Do not conclude from this backtest that HRP is the best method.
4. **ERC LW $\approx$ ERC**: with 10 assets and 252 days ($q \approx 0.04$), shrinkage is small (median δ^* of 4%, 15% at most). It becomes decisive from about fifty assets.
5. **HRP's hidden cost is its turnover**, 2.5 times that of ERC.

Average capital allocation
Mean target weight across rebalancings

	SPY	EFA	EEM	IWM	TLT	IEF	LQD	GLD	DBC	VNQ
1/N	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
InvVol	8%	8%	6%	6%	10%	21%	19%	9%	7%	6%
RB bond tilt	6%	6%	4%	5%	14%	29%	18%	7%	7%	4%
ERC	7%	7%	5%	6%	13%	26%	14%	9%	8%	5%
ERC LW	7%	7%	5%	6%	14%	24%	14%	10%	8%	5%
HRP	3%	4%	3%	2%	6%	39%	28%	7%	4%	3%

Figure 12. Average capital allocation.

Average risk allocation
Mean relative risk contribution, in-sample covariance at each rebalancing

	SPY	EFA	EEM	IWM	TLT	IEF	LQD	GLD	DBC	VNQ
1/N	13%	14%	18%	17%	0%	0%	3%	6%	11%	17%
InvVol	12%	13%	13%	12%	4%	5%	11%	8%	9%	13%
RB bond tilt	8%	8%	8%	8%	15%	15%	15%	8%	8%	8%
ERC	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
ERC LW	10%	10%	10%	10%	10%	9%	10%	10%	10%	10%
HRP	2%	4%	4%	2%	10%	31%	30%	10%	3%	5%

Figure 13. Average risk allocation (ex ante relative contributions).

These two tables summarize the difference between methods better than any metric.

- **1/N**: equal capital (10% each) but 79% of the risk on equity-like assets (SPY, EFA, EEM, IWM, VNQ). Treasuries contribute 0%, or even negatively, thanks to their negative correlation.

- **ERC**: exactly equal risk (10% each), at the cost of 39% of capital in IEF and TLT.
- **RB bond tilt**: the doubled budgets on TLT, IEF and LQD are met (15% each).
- **HRP**: 73% of capital and 71% of risk on the three bond ETFs, 67% of capital on IEF and LQD alone. The bisection treated the rates block as the other half of the universe and gave it massive weight as a low-volatility cluster.

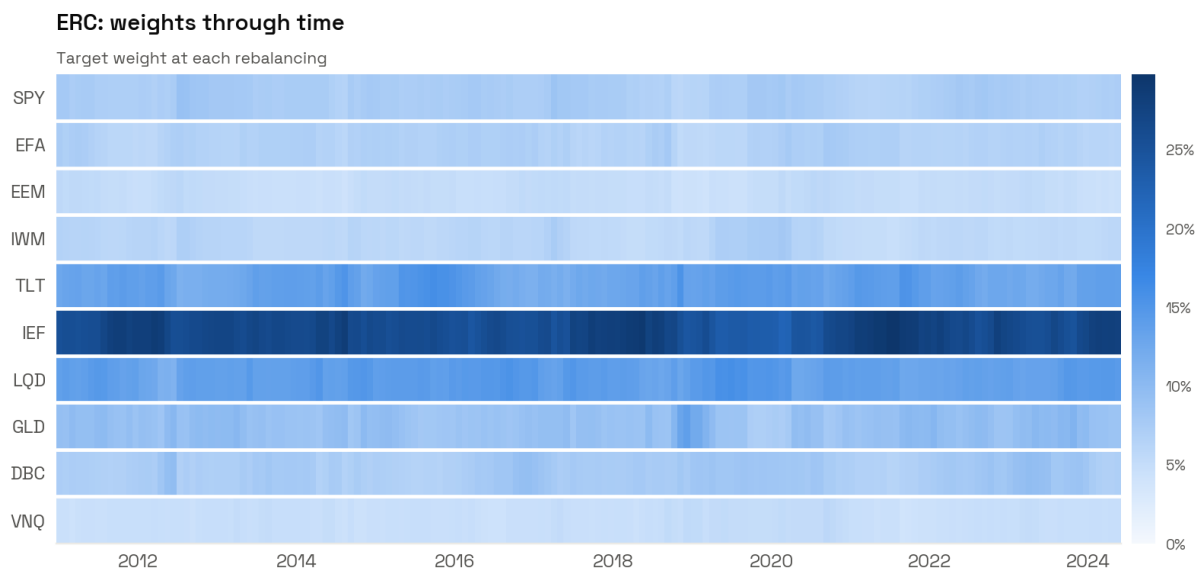


Figure 14. ERC weights across rebalancings.

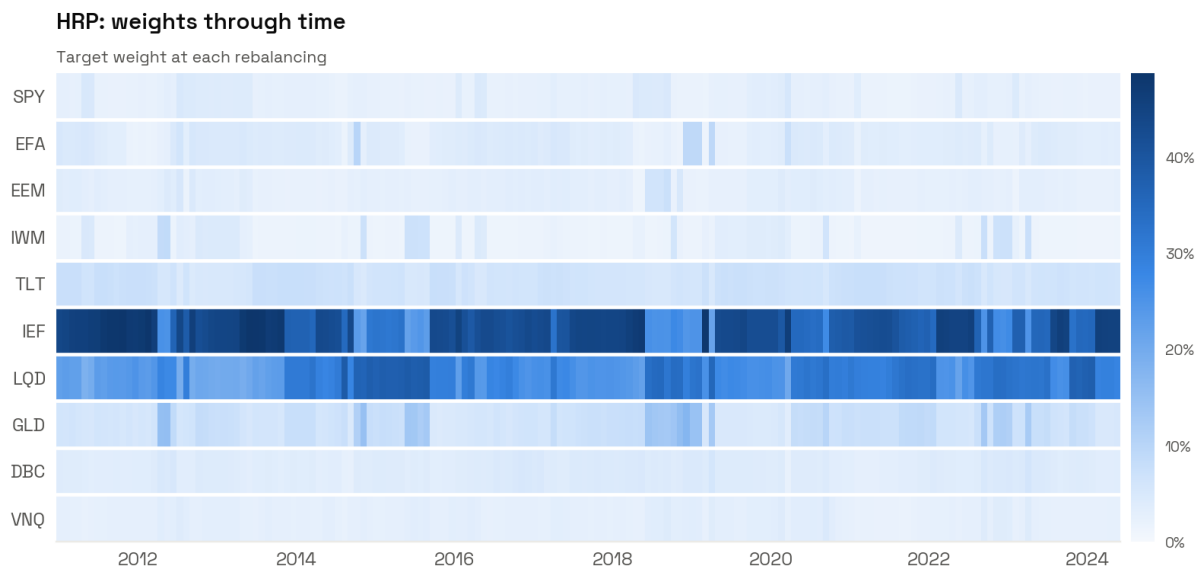


Figure 15. HRP weights across rebalancings.

ERC adjusts its weights continuously and smoothly. HRP jumps from one configuration to another whenever the clustering tree changes: this is the visible source of its turnover.

10 Expert checklist

Data

- ☐ Adjusted prices (total return), never returns on raw prices.
- ☐ No back-fill, no dropping of assets with a short history.
- ☐ Simple returns for the portfolio.

Estimation

- ☐ N/T ratio checked; above 0.1, shrinkage or a factor model.
- ☐ Expected returns never used without a strong prior or heavy shrinkage.

Backtest

- ☐ Weights computed with information up to $t - 1$ only.
- ☐ Weight drift modelled, turnover measured against drifted weights.
- ☐ Realistic costs (5 to 20 bps on liquid ETFs, more on small caps and emerging markets).
- ☐ Robustness tested on an (L, H) grid and on rebalancing offsets.
- ☐ 1/N benchmark always included.

Evaluation

- ☐ Comparison on Sharpe ratio, or at equal volatility, not on raw CAGR.
- ☐ Standard error of the Sharpe ratio computed before any conclusion.
- ☐ PSR for non-normality, DSR if several variants were tested.
- ☐ Drawdowns and CVaR examined, not just volatility.
- ☐ Risk contributions inspected: know where the risk is.
- ☐ Subperiods and regimes analysed: overall performance may come from a single episode.

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